INCH-POUND

A-A-59765

3 February 2005

COMMERCIAL ITEM DESCRIPTION

HELMET, NAVAL BATTLE (NBH)

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This commercial item description covers the requirements for a Naval Battle Helmet (NBH) to provide U.S. Navy personnel with head protection from ballistic fragments and impacts while performing duties on topside of ships.

2. SALIENT CHARACTERISTICS.

2.1 Physical and performance characteristics. The assembled Naval Battle Helmet shall be comprised of a helmet shell, an inner impact liner, an adjustable suspension frame assembly, and a chinstrap assembly. The helmet shall be configured to provide the user with cranial and ear protection from ballistic fragments and impacts. The helmet design shall incorporate a replaceable and adjustable suspension system including a ratchet-type headband for quick sizing adjustment. The helmet shall contain a chinstrap having a quick donning/doffing feature.

2.1.1 Ballistic resistance performance: The assembled helmet shall exhibit ballistic limits of not less than the limits given in Table I when tested using the following size steel fragments:

a. 4-grain RCC at 0 degree obliquity (see Figure 1)
b. 17-grain FSP at 0 degree obliquity (see Figure 2)
c. 17-grain FSP at 45 degrees obliquity (see Figure 2)
d. 64-grain RCC at 0 degree obliquity (see Figure 1)

2.1.1.1 Ballistic resistance test procedure. The ballistic resistance test shall be conducted in accordance with NATO/STANAG 2920 Ballistic Test Method for Personal Armours and as described in the following paragraphs. The helmets used during testing shall be stored in the ballistic test chamber for not less than 24 hours prior to testing.

2.1.1.2 Projectiles. Three different size projectiles shall be used during testing. The 4-grain and 64-grain projectiles shall conform to Figure 1 of this document. The 17-grain projectile shall be .22 caliber, and conform to Figure 2 of this document.

2.1.1.3 Barrel. The barrel shall be .22 caliber, rifled barrel for the 4-grain projectile, .30 caliber for the 17-grain projectile and .46 caliber for the 64-grain projectile. All barrels shall have a one-in-sixteen twist and a length of 28 inches. The barrel shall be chambered to accommodate firing the specified sabot.

2.1.1.4 Sabot. A sabot will be used in loading and firing the projectiles for the 4-grain, 17-grain and 64-grain projectiles, respectively.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05Q, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to commandstandards@navsea.navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2.1.1.5 Pusher plate. The pusher plate of the 4-grain projectile shall have a diameter of 0.22 inch and a thickness of 0.071 inch. The pusher plate for the 17-grain projectile shall have a diameter of 0.304 inch and a thickness of 0.050 inch. The pusher plate for the 64-grain projectile shall have a diameter of .508 inch and a thickness of 0.125 inch. The material shall be aluminum, 7075-T6 or 6061-T6.

2.1.1.6 Testing approach. The helmet shall be subdivided by marking the outside surface into five sections; a crown section and four side sections. The crown section shall be a 4-inch diameter circle. The four side sections shall be below the crown section and spaced at 90-degree intervals around the helmet. The side sections shall be identified by drawing lines from the center of the crown section to marks on the periphery of the helmet spaced every 90 degrees. For testing with 4-grain and 17-grain projectiles, a minimum of two impacts, randomly placed, shall be fired in each section, except that one of the two impacts in the crown section shall be at the approximate center of this section. The impacts shall be a minimum distance of 1½ inches from each other and from holes. For testing with 64-grain projectiles only one impact is required for each of the five sections. In all cases, the witness plate shall be rigidly mounted inside the helmet 2 inches behind the area of impact.

2.1.1.7 V_{50} ballistic limit, BL(P). The V_{50} ballistic limit [BL(P)] for impacts on each test helmet with 4-grain and 17-grain projectiles shall be the average of 6 fair impact velocities, consisting of the three lowest complete penetration velocities and the three highest partial penetration velocities, provided that the spread for the 6 velocities is not greater than the allowable ranges given in Table I. If the 6-round limit cannot be attained within the allowable range, the results are invalid. The V_{50} ballistic limit will be determined for each size fragment by averaging the V_{50} results for three test helmets. If either the average V_{50} for three helmets is below the value given in Table I or there is a complete penetration below the limit given in Table I, the results are invalid. The V_{50} BL(P) for 64-grain impacts on a test helmet shall be the average of the 4 impact velocities provided that two are partial penetrations and two are complete penetrations, and the spread is not greater than 125 ft/sec. If the limit cannot be attained within a velocity spread of 125 ft/sec or less, the results are invalid. Again, the V_{50} ballistic limit shall be determined by averaging the V_{50} results for three test helmets. If either the average V_{50} is below the given value in Table I or there is a complete penetration below the limit in Table I, the results are invalid.

2.1.2 Impact performance. The assembled helmet shall be tested for impact resistance in accordance with ANSI Z89.1. The helmet shall not transmit a force of more than 850 pounds during this test. For impact attenuation, the assembled helmet shall not transmit the impact energy to a standard head form accelerometer in excess of 300 g. The impact testing shall be in accordance with ANSI Z90.1, except that there will be a single side impact in four sites and each will be upon a flat steel anvil.

2.2 Design and construction.

2.2.1 Sizing. The helmet/impact liner shall be universal and fit head sizes from size 6¾ to size 8. A sketch of the Helmet Shell and Impact Liner is provided in Figure 3.

2.2.2 Helmet shell. The helmet shell shall be fabricated using a para-aramid ballistic material capable of meeting the performance requirements. A dimensional sketch of the helmet shell is provided in Figure 4. The weight of the unfinished helmet shell, not including external finish or edging, shall not exceed 39 ounces.

2.2.3 Coating. The external surface of the helmet shell shall be finished with an epoxy paint system consisting of a primer coat (color olive drab green), an aggregate coat (color olive drab green), which shall be 40/100 mesh walnut shell flour – 10 ounces of walnut flour to 1 gallon of coating, and a finish coat (color drab green) without aggregate.

2.2.4 Aggregate. The aggregate for shell texturing shall be silica sand or walnut shell flour and shall be incorporated in the second coat. This coat shall be applied to the outside surface of the helmet shell including the outside edge of the rubber edging.

2.2.5 Rubber edging. A rubber edging shall be applied to the edge of the helmet using a synthetic rubber base type adhesive. The edging shall be made of a rubber compound matching the outer shell and shall conform to the requirements of ASTM D1000 and ASTM D2000.

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2.2.6 Impact liner. The impact liner shall be molded of black expanded polypropylene material no greater than ⅝-inch thick. The outside contour of the liner shall match the inside contour of the helmet shell mold. The edge of the liner shall be 2⅜ inches from the bottom edge of the earflaps of the helmet shell. This height shall be maintained around the circumference of the liner to ensure that it is never exposed. The liner shall be secured to the helmet shell by means of hook and loop tape to permit removal. The hook and loop tape shall have a pressure sensitive adhesive adhered to the backside. Recesses in the liner shall be provided for the loop tape as well as the holding mechanism for the crown straps. A black nylon tricot material shall cover the edge of the liner to a minimum depth of one inch.

2.2.7 Adjustable suspension frame system. The design of each helmet shall incorporate an adjustable suspension frame system, which is removable and replaceable. This system shall include crown straps and a ratchet-type, quick-sizing headband which allows for adjustment from size 6¾ to size 8. The adjustable headband shall also permit a minimum of ½-inch vertical adjustment. The system shall have no metal springs. The adjustable headband shall be molded as one piece. The headband shall have a foam/cloth cover in front and back for comfort, and the crown straps shall be made of nylon webbing. A cushioned liner or padding shall be positioned at the crown strap intersection. The helmet suspension system shall be tested in accordance with NFPA 1972. The attachment points shall withstand a static load of 10 pounds without separation from the impact liner.

2.2.8 Chinstrap assembly. Each helmet shall have a chinstrap that consists of a ¾-inch wide aramid webbing with a high temperature resistant, quick release mechanism. The chinstrap shall have a webbed chin cup. The chinstrap shall be attached to the earflaps of the shell for a minimum of 1 inch from any surface. The helmet chinstrap shall be tested in accordance with NFPA 1972. The chinstrap shall withstand a static load of 100 pounds without breaking or separating from the helmet shell.

2.2.9 Labeling. There shall be a self-adhesive label affixed to the inside of the helmet shell and a second label affixed to the inside of the liner that indicates the Item Name, Manufacturer, NSN, and Navy Certification Number.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Market acceptability. Manufacturer shall show that they have made similar type helmets, which meet ballistic or National Institute of Justice requirements.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 ANSI. ANSI standards are available from the American National Standards Institute, 25 W. 43rd St, 4th Floor, New York, NY 10036 or online at http://webstore.ansi.org/.

6.1.2 ASTM. ASTM standards are available from ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959 or online at www.astm.org.

6.1.3 FAR. The Federal Acquisition Regulations (FAR) may be obtained from Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20401 or online at www.gpoaccess.gov/index.html.

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6.1.4 NATO/STANAG. North Atlantic Treaty Organization standardization agreements are available from NATO, Blvd. Leopold III 1110, Brussels, Belgium or online at www.nato.int/home.htm.

6.1.5 NFPA. National Fire Protection Association (NFPA) standards are available from NFPA, 1 Batterymarch Park, Quincy, MA 02169-7471 or online at www.nfpa.org.

6.2 Ordering data. The contract or order should specify the following:

a. CID document number and revision.
b. National stock numbers (NSNs).
c. Product conformance provisions.
d. Packaging requirements.

6.3 National stock numbers (NSNs). The following are the NSNs that are assigned that correspond to this CID. This list may not be indicative of all possible NSNs associated with the CID.

NSNs	Item
8470-01-455-3325	Helmet, Battle
8470-01-455-3338	Liner, Helmet
8470-01-455-3331	Shell, Helmet
8470-01-455-3349	Chin Strap Assembly

6.4 Testing laboratories. Commercial laboratories capable of performing the required ballistic testing are: H.P. White Laboratory, Bel Air, MD; Denver Research Institute, Denver, CO; and Gentex Corporation, Carbondale, PA.

6.5 Keywords.

Aramid
Ballistic
Fragmentation
Head protection
Impact acceleration
Impact liner
Impact resistance suspension frame assembly

MILITARY INTERESTS

Custodians:
Army – GL
Navy – SH
Air Force – 11

Review Activities:
Army – GL1, MR
Navy – CG, MC, NU
DLA – CT

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FSS

Preparing Activity:
Navy – SH
(Project 8470-0188-000)

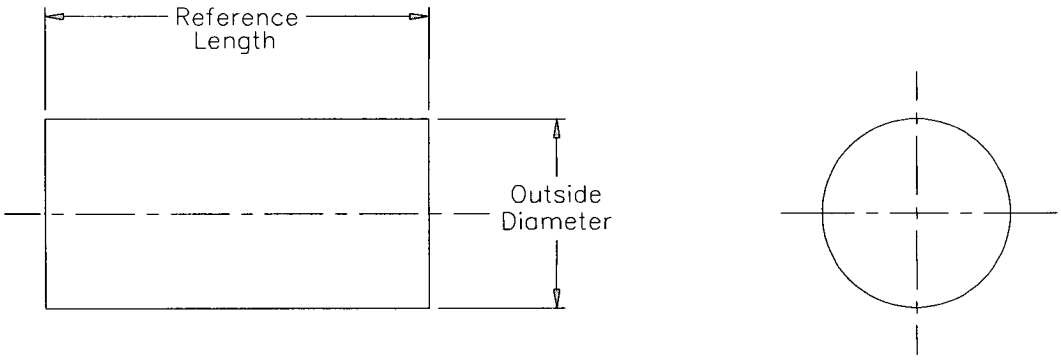
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TABLE I. Minimum ballistic resistance limits (ft/sec).

Fragment Size	Minimum V_{50}	Minimum Penetration Velocity	Allowable Velocity Range	Required Impacts
4-grain RCC	3100	3000	240	3 + 3
17-grain FSP (0 degrees)	2100	2000	150	3 + 3
17-grain FSP (45 degrees)	2300	2200	150	3 + 3
64-grain RCC	1600	1550	125	2 + 2



RCC Weight (Grains)	Outside Diameter (O.D.)	Reference Length (Inches)
4 ± 0.15	0.135 $^{+.001}_{-.000}$	0.141
64 ± 1	0.343 ±.001	0.350

FIGURE 1. Right circular cylinder (RCC) projectiles.

Notes:

1. Adjust Reference Length to Meet RCC Weight (Grains)
2. Material is Tool Steel (Oil Hardened Drill Rod), Heat Treated to Obtain Rockwell C Hardness of 29 ± 2.
3. Finish is 32√

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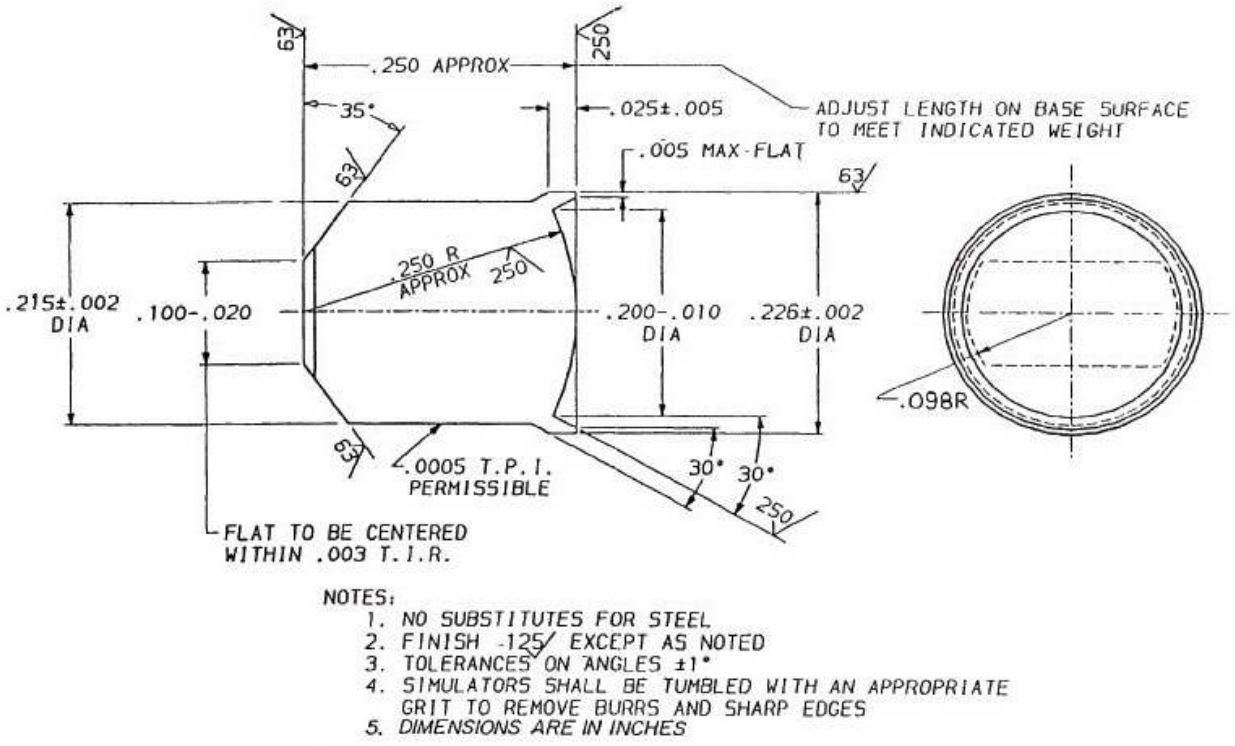


FIGURE 2. Fragment 17-grain.



FIGURE 3. Shell and impact liner view.

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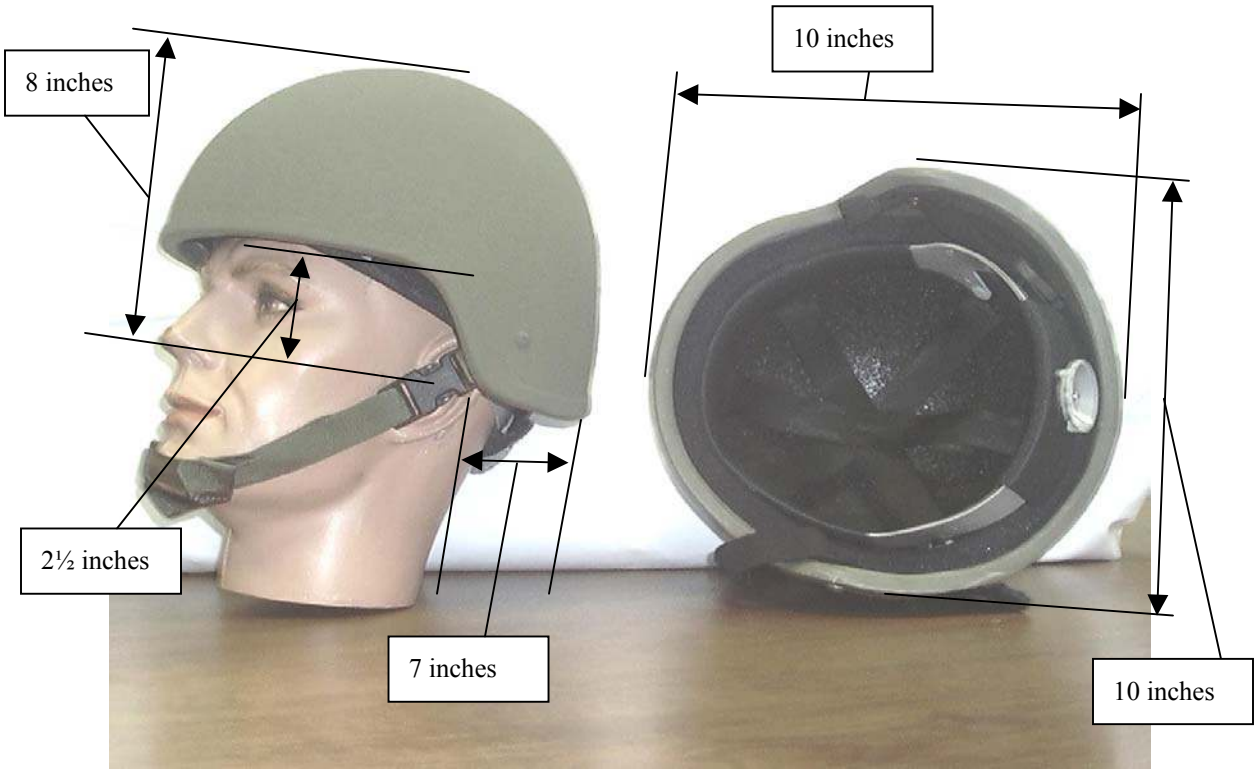


FIGURE 4. Approximate dimensions (inches).

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